

CURIONEST

CHEMISTRY FOUNDATIONS

MATH AND MEASUREMENT

Complete Instructional Unit for U.S. Grades 9-10

TEACH -> MODEL -> PRACTICE -> ASSESS

- 15 projectable teaching slides
- 12 student lesson, practice, and review pages
- Unit Tests A and B with 20 multiple-choice items each
- Teacher guide, full answers, rationales, and repair moves



Build the measurement habits students need before chemistry calculations begin.

5 LESSONS · TWO 20-ITEM TEST FORMS · PDF · US LETTER



UNIT PURPOSE

Students learn how measurements become defensible evidence. The unit moves from tools, units, and notation to factor-label reasoning, density, percent error, and evidence-based claims about accuracy and precision.

INSTRUCTIONAL RHYTHM

Engage -> Teach -> Model -> Guided Practice -> Independent Practice -> Exit Ticket -> Mixed Review -> Unit Test

SUGGESTED PACING

- 1 Lesson 1: Measurement Evidence and SI Records. Use Slides 1-3, Student Pages 1-2, and the exit ticket.
- 2 Lesson 2: Scientific Notation and Significant Figures. Model rounding only after the full calculation.
- 3 Lesson 3: Dimensional Analysis. Require units on every factor and a visible cancellation check.
- 4 Lesson 4: Density and Percent Error. Connect calculations to measured evidence and units.
- 5 Lesson 5: Accuracy, Precision, and Error. Compare repeated trials before making a claim.
- 6 Cumulative mixed review, feedback, and targeted reteaching.
- 7 Unit Test A. Reserve Form B for a retake, alternate class, or test security.

PACKAGE MAP

Projectable slides teach the lesson. Student pages provide guided notes, modeled thinking, supported practice, independent practice, and exit tickets. Tests A and B follow all instruction and review. The teacher file contains every answer and rationale.



Support, monitor, and respond

BEFORE TEACHING

Decide calculator expectations and the significant-figure convention used by your course. Confirm students can multiply powers of ten. Preview the pictured tools and remind students that the correct tool depends on the quantity and needed resolution.

FORMATIVE LOOK-FORS

Students attach units to measured values, distinguish exact conversion factors from measured data, cancel units before calculating, round once at the end, and use repeated trials when discussing precision.

SUPPORT

Keep the prefix table, factor-label frame, and formula bank visible. Let students annotate units in color. Reduce arithmetic load without removing the reasoning step.

EXTEND

Ask students to design a repeated-trial measurement plan, predict the effect of a systematic offset, and defend which additional data would strengthen an accuracy claim.

ASSESSMENT USE

Use Form A after instruction and mixed review. Reserve Form B for retakes, an alternate class period, or test security. Do not use either test as the first exposure to a skill.

SCOPE

Supports the NGSS practice of Using Mathematics and Computational Thinking; no formal performance-expectation alignment is claimed.

LEARNING TARGET

Distinguish qualitative and quantitative observations, choose an appropriate tool, and record a complete SI-style measurement.

NOTICE / PREDICT

Two students describe the same sample as 'heavy' and '18.62 g.' Which record can another scientist compare directly, and what makes it more useful?

Be ready to defend your first idea. You may revise it after the lesson.



Electronic balance: mass

Explain the idea

BIG IDEA

A measurement is evidence only when the value, unit, quantity, tool, and reasonable precision are communicated clearly.

ESSENTIAL TERMS

qualitative: descriptive evidence without a measured number

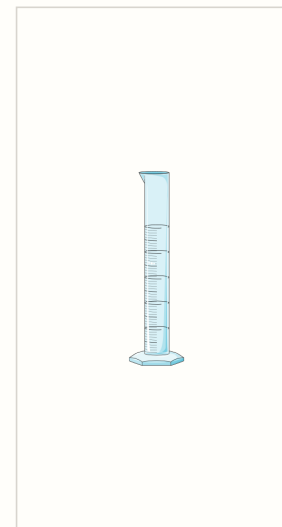
quantitative: evidence that includes a measured or counted number

resolution: the smallest increment an instrument can display or support

SI unit: a standard unit used to communicate a physical quantity



Electronic balance: mass



Graduated cylinder: liquid volume



Beaker: holding and approximate volume

WHAT TO REMEMBER

- A complete quantitative record names the measured quantity and gives a number with a unit.
- Choose the tool for the quantity and needed resolution: a balance measures mass; a graduated cylinder measures liquid volume more precisely than a beaker.
- Write a space between a value and its unit symbol, use correct capitalization, and do not pluralize symbols: 6.0 mL, not 6.0mls.
- A leading zero prevents a missed decimal: write 0.45 g, not .45 g.

Model the reasoning

WORKED EXAMPLE

A data table contains only '42.7.' Diagnose the record and repair it for a liquid-volume measurement.

- 1 Identify what is missing: the record does not name the measured quantity or unit.
- 2 Use the measurement context to choose the correct unit: milliliters for this liquid volume.
- 3 Preserve the reported precision and apply SI spacing and capitalization.

RESULT

Record the result as 'liquid volume = 42.7 mL' and identify the instrument used.

CHECK: Why is 24.6 mL more informative than the number 24.6 alone?

LEARNING TARGET

Represent measured values in scientific notation and apply significant-figure rules to calculations.

NOTICE / PREDICT

Two calculators display 39.68 and 39.680000. Does the longer display automatically make the experimental answer more precise?

Be ready to defend your first idea. You may revise it after the lesson.

Explain the idea

BIG IDEA

Scientific notation preserves place value; significant figures communicate justified precision from measured data.

ESSENTIAL TERMS

coefficient: the number from 1 up to, but not including, 10 in scientific notation

exponent: the power of ten that records decimal movement

significant figure: a digit justified by the measurement and its recorded precision

exact number: a counted or defined value that does not limit significant figures

WHAT TO REMEMBER

- Scientific notation has the form $a \times 10^n$, where $1 \leq a < 10$.
- Leading zeros are placeholders; zeros between nonzero digits and recorded trailing decimal zeros are significant.
- For multiplication and division, round to the fewest significant figures among measured values.
- For addition and subtraction, round to the least precise decimal place. Keep guard digits and round once at the end.

WORKED EXAMPLE

Calculate 12.4×3.20 and report the answer with justified precision.

1

Calculate without early rounding: $12.4 \times 3.20 = 39.68$.

2

Count significant figures: 12.4 has 3 and 3.20 has 3.

3

Round the product to 3 significant figures.

RESULT**39.7**

CHECK: Why is 39.68 not the final reported answer for the worked example?

LEARNING TARGET

Convert quantities by arranging conversion factors so unwanted units cancel and the target unit remains.

NOTICE / PREDICT

**A conversion gives the expected number,
but the unwanted unit never cancels.
Should the result be trusted?**

Be ready to defend your first idea. You may revise it after the lesson.



A rate such as km/h combines two units; both must be tracked during conversion.

BIG IDEA

A conversion factor equals one. Its orientation is chosen by units, and unit cancellation is the built-in reasonableness check.

ESSENTIAL TERMS

conversion factor: a ratio of equivalent quantities that equals one

factor-label method: a method that tracks and cancels units through multiplication

target unit: the unit required in the final answer

dimensional consistency: agreement between the units in a setup and the intended quantity



A rate such as km/h combines two units; both must be tracked during conversion.

WHAT TO REMEMBER

- Start with the given quantity and write its unit.
- Place each unwanted unit opposite itself so it cancels.
- Continue multiplying by equal ratios until only the target unit remains.
- Defined conversion factors, such as $1000\text{ m} = 1\text{ km}$, are exact and do not limit significant figures.

WORKED EXAMPLE

Convert 72.0 km/h to m/s.

1

Write $72.0 \text{ km/h} \times (1000 \text{ m} / 1 \text{ km})$ so kilometers cancel.

2

Multiply by $(1 \text{ h} / 3600 \text{ s})$ so hours cancel.

3

Calculate $72.0 \times 1000 / 3600$ and retain three significant figures.

RESULT**20.0 m/s**

CHECK: What does it mean if both kilometers and meters remain in the final unit?

LEARNING TARGET

Calculate density, rearrange the density relationship, and use percent error to compare an experimental result with an accepted value.

NOTICE / PREDICT

Two objects have different masses but the same density. What does density reveal that mass alone does not?

Be ready to defend your first idea. You may revise it after the lesson.



Read liquid volume at eye level; use the bottom of a concave meniscus.

BIG IDEA

Density relates mass to volume, while percent error measures the size of a difference relative to an accepted value.

ESSENTIAL TERMS

density: mass per unit volume, $d = m / V$

accepted value: the reference value used for comparison

experimental value: the value produced by a measurement or calculation

percent error: absolute difference divided by accepted value, multiplied by 100



Read liquid volume at eye level; use the bottom of a concave meniscus.

WHAT TO REMEMBER

- Density units combine mass and volume, commonly g/mL or g/cm³.
- Use $d = m / V$; rearrange to $m = dV$ or $V = m/d$ when needed.
- Percent error = $| \text{experimental} - \text{accepted} | / \text{accepted} \times 100\%$.
- Keep the absolute value in the numerator and use the accepted value as the denominator.

WORKED EXAMPLE

A sample has a mass of 57.0 g and a volume of 20.0 mL. Calculate its density.

- 1 Select $d = m / V$ and substitute measured values with units.
- 2 Calculate $57.0 \text{ g} / 20.0 \text{ mL} = 2.85 \text{ g/mL}$.
- 3 Both measurements have three significant figures, so report three.

RESULT**2.85 g/mL**

CHECK: Why must a density answer include both a mass unit and a volume unit?

LEARNING TARGET

Use repeated measurements and an accepted value to distinguish precision, accuracy, random variation, and systematic error.

NOTICE / PREDICT

Three measurements agree closely with one another but are all higher than the accepted value. Is the method precise, accurate, both, or neither?

Be ready to defend your first idea. You may revise it after the lesson.



A zero offset can create a systematic mass error.

BIG IDEA

Precision is agreement among repeated trials; accuracy is agreement with an accepted value. One statistic cannot establish both.

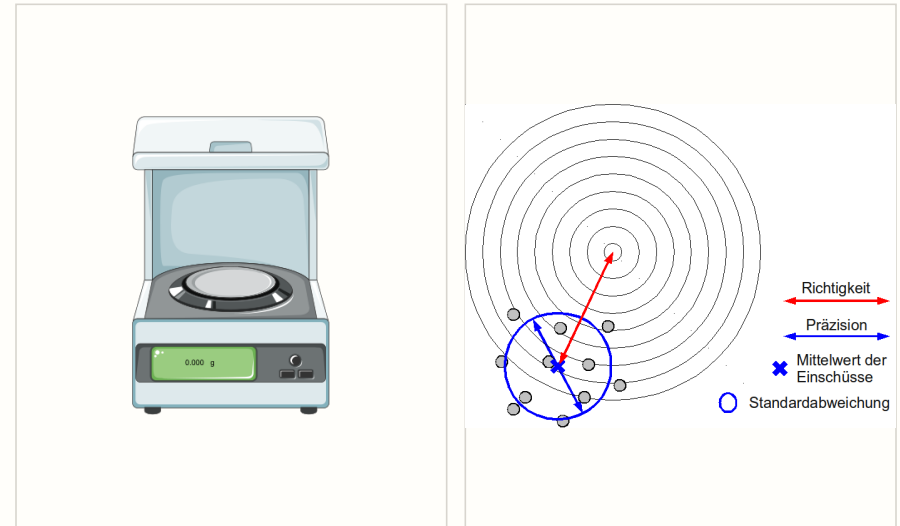
ESSENTIAL TERMS

precision: closeness of repeated measurements to one another

accuracy: closeness of a result to an accepted value

random error: unpredictable variation that changes from trial to trial

systematic error: a consistent offset caused by method or instrument bias



A zero offset can create a systematic mass error.

Center agreement and trial spread answer different questions.

WHAT TO REMEMBER

- Judge precision from the spread of repeated measurements, not from one trial.
- Judge accuracy by comparing a result or mean with an accepted value.
- Random variation increases scatter; repeated trials reveal its size.
- Systematic error shifts results in a consistent direction and is not fixed by averaging more biased trials.

WORKED EXAMPLE

Trials are 1.247, 1.249, and 1.248 g/mL; the accepted value is 1.200 g/mL. Evaluate the results.

1

Compare trials: the range is only 0.002 g/mL, so the measurements are precise.

2

Find the mean: $(1.247 + 1.249 + 1.248) / 3 = 1.248$ g/mL.

3

Compare with 1.200 g/mL: the mean is consistently high; percent error is 4.00%.

RESULT

Precise but not accurate relative to the accepted value; the pattern suggests a systematic high bias.

CHECK: Why would ten more trials not automatically remove a balance zero offset?

Learn and model

Name: _____ Date: _____ Period: _____

TARGET

Distinguish qualitative and quantitative observations, choose an appropriate tool, and record a complete SI-style measurement.

ENGAGE

Two students describe the same sample as 'heavy' and '18.62 g.' Which record can another scientist compare directly, and what makes it more useful?

TEACH

- A complete quantitative record names the measured quantity and gives a number with a unit.
- Choose the tool for the quantity and needed resolution: a balance measures mass; a graduated cylinder measures liquid volume more precisely than a beaker.
- Write a space between a value and its unit symbol, use correct capitalization, and do not pluralize symbols: 6.0 mL, not 6.0mls.
- A leading zero prevents a missed decimal: write 0.45 g, not .45 g.

TERMS

qualitative: descriptive evidence without a measured number

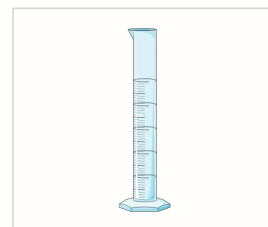
quantitative: evidence that includes a measured or counted number

resolution: the smallest increment an instrument can display or support

SI unit: a standard unit used to communicate a physical quantity



Electronic balance: mass



Graduated cylinder: liquid volume

WORKED EXAMPLE

A data table contains only '42.7.' Diagnose the record and repair it for a liquid-volume measurement.

1. Identify what is missing: the record does not name the measured quantity or unit.
2. Use the measurement context to choose the correct unit: milliliters for this liquid volume.
3. Preserve the reported precision and apply SI spacing and capitalization.

RESULT: Record the result as 'liquid volume = 42.7 mL' and identify the instrument used.

Practice and check

Name: _____ Date: _____ Period: _____

GUIDED PRACTICE

L1-1. Classify 'The solution is pale blue' and 'The solution volume is 18.5 mL' as qualitative or quantitative.

Scaffold: Look for a measured number and unit.

INDEPENDENT

L1-2. Choose the pictured tool best suited to measure 24.6 mL of water. Cite one visual feature.

INDEPENDENT

L1-3. Repair each record: (a) .45g (b) 6.0ml (c) 5 kgs.

INDEPENDENT

L1-4. A student needs the mass of a dry solid. Explain why a graduated cylinder is not the correct first tool.

EXIT TICKET

L1-5. Write one complete quantitative record for a sample with a mass display of 7.250 g.

Learn and model

Name: _____ Date: _____ Period: _____

TARGET

Represent measured values in scientific notation and apply significant-figure rules to calculations.

ENGAGE

Two calculators display 39.68 and 39.680000. Does the longer display automatically make the experimental answer more precise?

TEACH

- Scientific notation has the form $a \times 10^n$, where $1 \leq a < 10$.
- Leading zeros are placeholders; zeros between nonzero digits and recorded trailing decimal zeros are significant.
- For multiplication and division, round to the fewest significant figures among measured values.
- For addition and subtraction, round to the least precise decimal place. Keep guard digits and round once at the end.

TERMS

coefficient: the number from 1 up to, but not including, 10 in scientific notation

exponent: the power of ten that records decimal movement

significant figure: a digit justified by the measurement and its recorded precision

exact number: a counted or defined value that does not limit significant figures

WORKED EXAMPLE

Calculate 12.4×3.20 and report the answer with justified precision.

1. Calculate without early rounding: $12.4 \times 3.20 = 39.68$.

2. Count significant figures: 12.4 has 3 and 3.20 has 3.

3. Round the product to 3 significant figures.

RESULT: 39.7

Practice and check

Name: _____ Date: _____ Period: _____

GUIDED PRACTICE

L2-1. Write 0.0005070 in scientific notation and state its significant figures.

Scaffold: Move the decimal until the coefficient is between 1 and 10; preserve recorded trailing zeros.

INDEPENDENT

L2-2. State the number of significant figures in (a) 0.004060 (b) 700. (c) 1.0200 (d) 0.030.

INDEPENDENT

L2-3. Calculate $45.60 / 2.5$ and round correctly.

INDEPENDENT

L2-4. Calculate $18.42 + 3.7 + 0.006$ and round correctly.

EXIT TICKET

L2-5. A student rounds every intermediate step. State one risk and one repair.

Learn and model

Name: _____ Date: _____ Period: _____

TARGET

Convert quantities by arranging conversion factors so unwanted units cancel and the target unit remains.

ENGAGE

A conversion gives the expected number, but the unwanted unit never cancels. Should the result be trusted?

TEACH

- Start with the given quantity and write its unit.
- Place each unwanted unit opposite itself so it cancels.
- Continue multiplying by equal ratios until only the target unit remains.
- Defined conversion factors, such as $1000\text{ m} = 1\text{ km}$, are exact and do not limit significant figures.

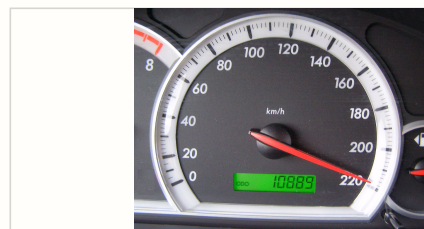
TERMS

conversion factor: a ratio of equivalent quantities that equals one

factor-label method: a method that tracks and cancels units through multiplication

target unit: the unit required in the final answer

dimensional consistency: agreement between the units in a setup and the intended quantity



A rate such as km/h combines two units; both must be tracked during conversion.

WORKED EXAMPLE

Convert 72.0 km/h to m/s.

1. Write $72.0\text{ km/h} \times (1000\text{ m} / 1\text{ km})$ so kilometers cancel.
2. Multiply by $(1\text{ h} / 3600\text{ s})$ so hours cancel.
3. Calculate $72.0 \times 1000 / 3600$ and retain three significant figures.

RESULT: 20.0 m/s

Practice and check

Name: _____ Date: _____ Period: _____

GUIDED PRACTICE

L3-1. Convert 3.45 km to meters using a visible conversion factor.

Scaffold: Place kilometers in the denominator of the factor so they cancel.

INDEPENDENT

L3-2. Convert 0.0825 L to milliliters.

INDEPENDENT

L3-3. Convert 2.60×10^5 mg to grams and preserve significant figures.

INDEPENDENT

L3-4. Convert 18.0 m/s to km/h.

EXIT TICKET

L3-5. Explain why a setup that ends in h/s is incomplete when the target is m/s.

Learn and model

Name: _____ Date: _____ Period: _____

TARGET

Calculate density, rearrange the density relationship, and use percent error to compare an experimental result with an accepted value.

ENGAGE

Two objects have different masses but the same density. What does density reveal that mass alone does not?

TEACH

- Density units combine mass and volume, commonly g/mL or g/cm³.
- Use $d = m / V$; rearrange to $m = dV$ or $V = m/d$ when needed.
- Percent error = $|\text{experimental} - \text{accepted}| / \text{accepted} \times 100\%$.
- Keep the absolute value in the numerator and use the accepted value as the denominator.

TERMS

density: mass per unit volume, $d = m / V$

accepted value: the reference value used for comparison

experimental value: the value produced by a measurement or calculation

percent error: absolute difference divided by accepted value, multiplied by 100



Read liquid volume at eye level; use the bottom of a concave meniscus.

WORKED EXAMPLE

A sample has a mass of 57.0 g and a volume of 20.0 mL. Calculate its density.

1. Select $d = m / V$ and substitute measured values with units.
2. Calculate $57.0 \text{ g} / 20.0 \text{ mL} = 2.85 \text{ g/mL}$.
3. Both measurements have three significant figures, so report three.

RESULT: 2.85 g/mL



Density and Percent Error

Practice and check

Name: _____ Date: _____ Period: _____

GUIDED PRACTICE

L4-1. A 31.6 g sample occupies 25.5 mL. Calculate density.

Scaffold: Write $d = m / V$ before substituting values.

INDEPENDENT

L4-2. A liquid has density 1.24 g/mL and volume 25.5 mL. Calculate mass.

INDEPENDENT

L4-3. A solid has mass 14.9 g and density 2.50 g/cm³. Calculate volume.

INDEPENDENT

L4-4. An experimental value is 12.4 g/mL and the accepted value is 12.0 g/mL. Calculate percent error.

EXIT TICKET

L4-5. Explain why percent error uses the accepted value in the denominator.

Learn and model

Name: _____ Date: _____ Period: _____

TARGET

Use repeated measurements and an accepted value to distinguish precision, accuracy, random variation, and systematic error.

ENGAGE

Three measurements agree closely with one another but are all higher than the accepted value. Is the method precise, accurate, both, or neither?

TEACH

- Judge precision from the spread of repeated measurements, not from one trial.
- Judge accuracy by comparing a result or mean with an accepted value.
- Random variation increases scatter; repeated trials reveal its size.
- Systematic error shifts results in a consistent direction and is not fixed by averaging more biased trials.

TERMS

precision: closeness of repeated measurements to one another

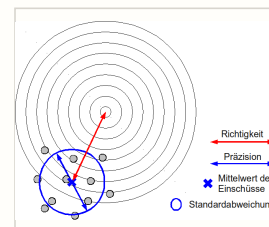
accuracy: closeness of a result to an accepted value

random error: unpredictable variation that changes from trial to trial

systematic error: a consistent offset caused by method or instrument bias



A zero offset can create a systematic mass error.



Center agreement and trial spread answer different questions.

WORKED EXAMPLE

Trials are 1.247, 1.249, and 1.248 g/mL; the accepted value is 1.200 g/mL. Evaluate the results.

1. Compare trials: the range is only 0.002 g/mL, so the measurements are precise.
2. Find the mean: $(1.247 + 1.249 + 1.248) / 3 = 1.248$ g/mL.
3. Compare with 1.200 g/mL: the mean is consistently high; percent error is 4.00%.

RESULT: Precise but not accurate relative to the accepted value; the pattern suggests a systematic high bias.

Practice and check

Name: _____ Date: _____ Period: _____

GUIDED PRACTICE

L5-1. Trials are 10.01, 10.00, and 10.02 g; accepted mass is 10.01 g. Classify accuracy and precision.

Scaffold: First compare the trials with one another; then compare their center with the accepted value.

INDEPENDENT

L5-2. Trials are 8.42, 8.43, and 8.42 mL; accepted volume is 8.00 mL. Classify accuracy and precision.

INDEPENDENT

L5-3. A student reads a meniscus from above. Name the error and give a specific repair.

INDEPENDENT

L5-4. A balance reads +0.20 g when empty. Explain why repeating trials without zeroing it does not solve the problem.

EXIT TICKET

L5-5. Explain why percent error alone cannot establish precision.



Cumulative Mixed Review

STUDENT REVIEW

Use ideas from every lesson

Name: _____ Date: _____ Period: _____

R1. Rewrite '.075kgs' as a correct SI-style record.

R2. Write 0.0008200 in scientific notation and state its significant figures.

R3. Calculate 6.20×4.1 and report the correct significant figures.

R4. Calculate $12.08 + 0.6 + 3.214$ and apply the decimal-place rule.

R5. Convert 2.75 h to seconds using dimensional analysis.



Cumulative Mixed Review

Use ideas from every lesson

Name: _____ Date: _____ Period: _____

R6. A sample has mass 24.96 g and volume 10.00 mL. Calculate density.

R7. A material has density 3.60 g/mL and volume 7.25 mL. Calculate mass.

R8. Experimental density is 2.40 g/mL; accepted density is 2.50 g/mL. Calculate percent error.

R9. Trials 24.9, 25.0, and 25.1 g surround an accepted value of 25.0 g. Evaluate accuracy and precision.

R10. A conversion result has the unit kg·mL/g when the target is mL. Diagnose the setup.



Unit Test - Form A

Math and Measurement

STUDENT ASSESSMENT

Name: _____ Date: _____ Period: _____

Directions: Select the best answer for each question. Mark one choice only. Complete this test after all lessons and the cumulative review.

1. Which observation is quantitative?

- A. The solid is shiny.
- B. The sample mass is 12.40 g.
- C. The vapor has a sharp odor.
- D. The liquid appears clear.

2. Which tool is best for measuring 24.6 mL of water?

- A. Electronic balance
- B. Beaker
- C. Graduated cylinder
- D. Thermometer

3. Which record follows SI writing conventions?

- A. 0.45 g
- B. .45g
- C. 0.45 gs
- D. 0.45 G

4. What is 0.0825 L in milliliters?

- A. 0.0000825 mL
- B. 0.825 mL
- C. 8.25 mL
- D. 82.5 mL

5. Which is 0.0005070 in scientific notation?

- A. 5.07×10^{-3}
- B. 5.070×10^{-4}
- C. 50.70×10^{-5}
- D. 0.5070×10^{-3}



Unit Test - Form A

Math and Measurement

STUDENT ASSESSMENT

Name: _____ Date: _____ Period: _____

6. How many significant figures are in 700.?

- A. 3
- B. 1
- C. 2
- D. 0

7. What is 12.4×3.20 with correct significant figures?

- A. 39.68
- B. 40
- C. 39.7
- D. 39.680

8. What is $18.42 + 3.7 + 0.006$ with correct precision?

- A. 22.126
- B. 22.13
- C. 22
- D. 22.1

9. Which factor converts 2.50 h directly toward seconds?

- A. $3600 \text{ s} / 1 \text{ h}$
- B. $1 \text{ h} / 3600 \text{ s}$
- C. $60 \text{ h} / 1 \text{ min}$
- D. $1 \text{ s} / 3600 \text{ h}$

10. What is 90.0 km/h in m/s?

- A. 2.50 m/s
- B. 9.00 m/s
- C. 25.0 m/s
- D. 32.4 m/s



Unit Test - Form A

Math and Measurement

STUDENT ASSESSMENT

Name: _____ Date: _____ Period: _____

11. A sample has mass 57.0 g and volume 20.0 mL. What is its density?

- A. 0.351 g/mL
- B. 2.85 g/mL
- C. 37.0 g/mL
- D. 1140 g/mL

12. A liquid has density 1.24 g/mL and volume 25.5 mL. What is its mass?

- A. 0.0486 g
- B. 20.6 g
- C. 30.74 g
- D. 31.6 g

13. Experimental density is 12.4 g/mL; accepted density is 12.0 g/mL. What is percent error?

- A. 0.0333%
- B. 0.400%
- C. 3.33%
- D. 96.8%

14. Trials are 5.01, 5.00, and 5.02 g. Which claim is supported without an accepted value?

- A. The trials are precise.
- B. The trials are accurate.
- C. The method has zero error.
- D. The balance is calibrated.

15. A mean is very close to an accepted value. Which property is most directly supported?

- A. High resolution
- B. High precision
- C. Random error
- D. Accuracy



Unit Test - Form A

Math and Measurement

STUDENT ASSESSMENT

Name: _____ Date: _____ Period: _____

16. What is the best repair for reading a concave meniscus from above?

- A. Use a wider container.
- B. Read at eye level at the bottom of the meniscus.
- C. Add more liquid.
- D. Round to fewer digits.

17. Why can percent error not establish precision by itself?

- A. Precision requires repeated trials.
- B. Percent error has no units.
- C. Accepted values are always exact.
- D. Precision uses only qualitative data.

18. A conversion ends with both km and m in the units. What should the student do?

- A. Round the number.
- B. Add the units.
- C. Ignore one unit.
- D. Reorient a factor so the unwanted unit cancels.

19. Samples of 5.0 g and 50.0 g of the same material have the same density. What does this show?

- A. Density is exact.
- B. Mass is intensive.
- C. Density is intensive.
- D. Volume is constant.

20. Which final record communicates a calculated density most completely?

- A. 2.85
- B. 2.85 g/mL
- C. about three
- D. density was measured



Unit Test - Form B

Math and Measurement

STUDENT ASSESSMENT

Name: _____ Date: _____ Period: _____

Directions: Select the best answer for each question. Mark one choice only. Complete this test after all lessons and the cumulative review.

1. Which observation is qualitative?

- A. The crystal is colorless.
- B. The temperature is 22.4 °C.
- C. The mass is 6.250 g.
- D. The volume is 15.0 mL.

2. Which tool should be used first to measure the mass of a dry solid?

- A. Graduated cylinder
- B. Electronic balance
- C. Beaker
- D. Meter stick

3. Which record uses the correct symbol and spacing for milliliters?

- A. 6.0ml
- B. 6.0 mls
- C. 6.0 ML
- D. 6.0 mL

4. Treating 7400 micrograms as two significant figures, what is the mass in milligrams?

- A. 0.0074 mg
- B. 0.74 mg
- C. 7.4 mg
- D. 74 mg

5. What is 8.12×10^5 in standard notation?

- A. 0.000812
- B. 81,200
- C. 812,000
- D. 8,120,000



Unit Test - Form B

Math and Measurement

STUDENT ASSESSMENT

Name: _____ Date: _____ Period: _____

6. How many significant figures are in 1.0200?

- A. 3
- B. 4
- C. 2
- D. 5

7. What is $45.60 / 2.5$ with correct significant figures?

- A. 18
- B. 18.2
- C. 18.24
- D. 20

8. What is $100.0 - 0.37$ with correct decimal-place precision?

- A. 99.63
- B. 99.6
- C. 100
- D. 99.630

9. What is 3.60 days in hours?

- A. 0.150 h
- B. 15.0 h
- C. 72.0 h
- D. 86.4 h

10. What is 18.0 m/s in km/h?

- A. 5.00 km/h
- B. 64.8 km/h
- C. 180 km/h
- D. 648 km/h



Unit Test - Form B

Math and Measurement

STUDENT ASSESSMENT

Name: _____ Date: _____ Period: _____

11. A sample has mass 44.7 g and volume 15.0 mL. What is its density?

- A. 0.336 g/mL
- B. 2.68 g/mL
- C. 2.98 g/mL
- D. 670 g/mL

12. A solid has mass 14.9 g and density 2.50 g/cm³. What is its volume?

- A. 5.96 cm³
- B. 12.4 cm³
- C. 17.4 cm³
- D. 37.3 cm³

13. Experimental mass is 24.0 g; accepted mass is 25.0 g. What is percent error?

- A. 1.00%
- B. 4.00%
- C. 24.0%
- D. 96.0%

14. Measurements are spread out but their mean equals the accepted value. Which description fits best?

- A. Precise and accurate
- B. Precise but not accurate
- C. Neither precise nor accurate
- D. Accurate on average but not precise

15. Measurements cluster tightly above the accepted value. Which description fits best?

- A. Accurate but not precise
- B. Precise but not accurate
- C. Both accurate and precise
- D. Neither can be judged



Unit Test - Form B

Math and Measurement

STUDENT ASSESSMENT

Name: _____ Date: _____ Period: _____

16. An empty balance reads +0.20 g. What type of problem is this?

- A. A systematic offset
- B. Random variation
- C. A significant-figure rule
- D. A qualitative observation

17. Why are repeated trials useful?

- A. They guarantee accuracy.
- B. They remove every bias.
- C. They reveal the spread and support a precision judgment.
- D. They make units unnecessary.

18. Why does the exact factor $1000\text{ m} = 1\text{ km}$ not limit significant figures?

- A. It is a defined relationship.
- B. It is larger than one.
- C. It contains a zero.
- D. It is written as a fraction.

19. Compared with 12.3 mL, what does 12.30 mL communicate?

- A. A larger volume
- B. A different unit
- C. A less accurate tool
- D. One additional recorded decimal place

20. Which answer is complete for a calculated volume?

- A. 5.96 cm^3
- B. 5.96
- C. volume is small
- D. about six

L1-1. Classify 'The solution is pale blue' and 'The solution volume is 18.5 mL' as qualitative or quantitative.

ANSWER: 'Pale blue' is qualitative. '18.5 mL' is quantitative because it includes a measured value and unit.

L1-2. Choose the pictured tool best suited to measure 24.6 mL of water. Cite one visual feature.

ANSWER: The graduated cylinder; its narrow body and graduated scale support a more precise volume reading than a beaker.

L1-3. Repair each record: (a) .45g (b) 6.0ml (c) 5 kgs.

ANSWER: (a) 0.45 g (b) 6.0 mL (c) 5 kg.

L1-4. A student needs the mass of a dry solid. Explain why a graduated cylinder is not the correct first tool.

ANSWER: A graduated cylinder measures volume. An electronic balance directly measures mass.

L1-5. Write one complete quantitative record for a sample with a mass display of 7.250 g.

ANSWER: Example: sample mass = 7.250 g, measured with an electronic balance.

MISCONCEPTION

A number automatically makes an observation scientific and complete.

REPAIR

Require students to underline the value once and circle the unit; if either mark is missing, the record is incomplete.

L2-1. Write 0.0005070 in scientific notation and state its significant figures.

ANSWER: 5.070×10^{-4} ; four significant figures.

L2-2. State the number of significant figures in (a) 0.004060 (b) 700. (c) 1.0200 (d) 0.030.

ANSWER: (a) 4 (b) 3 (c) 5 (d) 2.

L2-3. Calculate $45.60 / 2.5$ and round correctly.

ANSWER: 18; the calculator result 18.24 is rounded to two significant figures.

L2-4. Calculate $18.42 + 3.7 + 0.006$ and round correctly.

ANSWER: 22.1; the tenths place limits the sum.

L2-5. A student rounds every intermediate step. State one risk and one repair.

ANSWER: Early rounding can accumulate error. Keep guard digits and round once after the final operation.

MISCONCEPTION

All zeros are either always significant or never significant.

REPAIR

Have students label each zero as leading, captive, or trailing before counting significant figures.

L3-1. Convert 3.45 km to meters using a visible conversion factor.

ANSWER: $3.45 \text{ km} \times (1000 \text{ m} / 1 \text{ km}) = 3.45 \times 10^3 \text{ m}$.

L3-2. Convert 0.0825 L to milliliters.

ANSWER: 82.5 mL.

L3-3. Convert $2.60 \times 10^5 \text{ mg}$ to grams and preserve significant figures.

ANSWER: $2.60 \times 10^2 \text{ g}$, or 260. g.

L3-4. Convert 18.0 m/s to km/h.

ANSWER: 64.8 km/h.

L3-5. Explain why a setup that ends in h/s is incomplete when the target is m/s.

ANSWER: The distance unit was not converted or canceled correctly; dimensional consistency requires only m/s to remain.

MISCONCEPTION

Conversion means moving a decimal according to a memorized direction.

REPAIR

Cover the numbers and inspect only the units. Reorient factors until every unwanted unit cancels.

Practice answers and repair moves

L4-1. A 31.6 g sample occupies 25.5 mL. Calculate density.

ANSWER: $31.6 \text{ g} / 25.5 \text{ mL} = 1.24 \text{ g/mL}$.

L4-2. A liquid has density 1.24 g/mL and volume 25.5 mL. Calculate mass.

ANSWER: $m = dV = 31.6 \text{ g}$.

L4-3. A solid has mass 14.9 g and density 2.50 g/cm³. Calculate volume.

ANSWER: $V = m/d = 5.96 \text{ cm}^3$.

L4-4. An experimental value is 12.4 g/mL and the accepted value is 12.0 g/mL. Calculate percent error.

ANSWER: $|12.4 - 12.0| / 12.0 \times 100 = 3.33\%$.

L4-5. Explain why percent error uses the accepted value in the denominator.

ANSWER: The accepted value provides the reference scale, so the difference is expressed relative to what is expected.

MISCONCEPTION

Density is the same as mass, or a larger sample must have a larger density.

REPAIR

Compare two amounts of the same material and calculate m/V for each; the ratio stays constant while mass and volume change.

L5-1. Trials are 10.01, 10.00, and 10.02 g; accepted mass is 10.01 g. Classify accuracy and precision.

ANSWER: The trials are precise and accurate: they cluster tightly and center on the accepted value.

L5-2. Trials are 8.42, 8.43, and 8.42 mL; accepted volume is 8.00 mL. Classify accuracy and precision.

ANSWER: Precise but not accurate relative to 8.00 mL.

L5-3. A student reads a meniscus from above. Name the error and give a specific repair.

ANSWER: Parallax/line-of-sight error. Move the eye to liquid level and read the bottom of the concave meniscus.

L5-4. A balance reads +0.20 g when empty. Explain why repeating trials without zeroing it does not solve the problem.

ANSWER: The zero offset is systematic; every result is shifted high by about 0.20 g. Zero or calibrate the balance first.

L5-5. Explain why percent error alone cannot establish precision.

ANSWER: Percent error compares with an accepted value; precision requires repeated trials to judge their agreement.

MISCONCEPTION

A low percent error proves the procedure is precise.

REPAIR

Ask for the repeated-trial data. If no spread can be examined, a precision claim is unsupported.



Cumulative Review Key

Answers from all lessons

TEACHER KEY

R1. Rewrite '.075kgs' as a correct SI-style record.

ANSWER: 0.075 kg.

R2. Write 0.0008200 in scientific notation and state its significant figures.

ANSWER: 8.200×10^{-4} ; four significant figures.

R3. Calculate 6.20×4.1 and report the correct significant figures.

ANSWER: 25; the product 25.42 rounds to two significant figures.

R4. Calculate $12.08 + 0.6 + 3.214$ and apply the decimal-place rule.

ANSWER: 15.9.

R5. Convert 2.75 h to seconds using dimensional analysis.

ANSWER: $2.75 \text{ h} \times 60 \text{ min/h} \times 60 \text{ s/min} = 9.90 \times 10^3 \text{ s}.$



Cumulative Review Key

Answers from all lessons

TEACHER KEY

R6. A sample has mass 24.96 g and volume 10.00 mL. Calculate density.

ANSWER: 2.496 g/mL.

R7. A material has density 3.60 g/mL and volume 7.25 mL. Calculate mass.

ANSWER: 26.1 g.

R8. Experimental density is 2.40 g/mL; accepted density is 2.50 g/mL. Calculate percent error.

ANSWER: 4.00%.

R9. Trials 24.9, 25.0, and 25.1 g surround an accepted value of 25.0 g. Evaluate accuracy and precision.

ANSWER: Precise and accurate: trials cluster tightly and center on the accepted value.

R10. A conversion result has the unit kg·mL/g when the target is mL. Diagnose the setup.

ANSWER: An unwanted mass unit did not cancel; reverse or replace the mass conversion factor before calculating.



Unit Test A - Rationale Key

Ten items per page

TEACHER KEY

1. B - The sample mass is 12.40 g.

A quantitative observation includes a measured number and unit: 12.40 g.

2. C - Graduated cylinder

A graduated cylinder is designed for liquid-volume measurement and supports finer readings than a beaker.

3. A - 0.45 g

The record uses a leading zero, a space, and the correct lowercase unit symbol without a plural s.

4. D - 82.5 mL

Multiply liters by the exact factor 1000 mL/L: $0.0825 \text{ L} = 82.5 \text{ mL}$.

5. B - 5.070×10^{-4}

A normalized coefficient lies between 1 and 10, and the recorded trailing zero preserves four significant figures.

6. A - 3

The decimal point shows that both trailing zeros are recorded as significant, giving three significant figures.

7. C - 39.7

Both factors have three significant figures, so 39.68 rounds to 39.7.

8. D - 22.1

The tenths place is least precise, so the sum 22.126 rounds to 22.1.

9. A - 3600 s / 1 h

Hours must cancel, so hours belong in the denominator and seconds in the numerator.

10. C - 25.0 m/s

$90.0 \times 1000 / 3600 = 25.0 \text{ m/s}$, retaining three significant figures.



Unit Test A - Rationale Key

Ten items per page

TEACHER KEY

11. B - 2.85 g/mL

Density is mass divided by volume: $57.0 / 20.0 = 2.85 \text{ g/mL}$.

12. D - 31.6 g

$m = dV = 1.24 \times 25.5 = 31.62 \text{ g}$, which rounds to 31.6 g.

13. C - 3.33%

$|12.4 - 12.0| / 12.0 \times 100 = 3.33\%$.

14. A - The trials are precise.

The tight grouping supports precision. Accuracy requires comparison with an accepted value.

15. D - Accuracy

Accuracy describes agreement with an accepted value.

16. B - Read at eye level at the bottom of the meniscus.

Eye-level viewing at the bottom of a concave meniscus reduces parallax error.

17. A - Precision requires repeated trials.

Precision is agreement among repeated measurements, so a single percent-error value is insufficient.

18. D - Reorient a factor so the unwanted unit cancels.

Uncanceled units reveal an incomplete setup; conversion factors must be oriented by unit cancellation.

19. C - Density is intensive.

Density does not depend on sample amount under the same conditions, so it is intensive.

20. B - 2.85 g/mL

The value includes a justified numerical result and the compound unit required for density.

1. A - The crystal is colorless.

Colorless is a descriptive observation without a measured number.

2. B - Electronic balance

An electronic balance directly measures mass.

3. D - 6.0 mL

Milliliter uses lowercase m, uppercase L, a space after the value, and no plural s.

4. C - 7.4 mg

1000 micrograms = 1 milligram, so 7400 micrograms = 7.4 mg.

5. C - 812,000

A positive exponent of 5 moves the decimal five places right: 812,000.

6. D - 5

The captive zero and both trailing decimal zeros are significant, for five total.

7. A - 18

The divisor has two significant figures, so 18.24 rounds to 18.

8. B - 99.6

100.0 is precise to the tenths place, so 99.63 rounds to 99.6.

9. D - 86.4 h

$3.60 \text{ days} \times 24 \text{ h/day} = 86.4 \text{ h}$; 24 is an exact defined factor.

10. B - 64.8 km/h

$18.0 \times 3600 / 1000 = 64.8 \text{ km/h}$.



Unit Test B - Rationale Key

Ten items per page

TEACHER KEY

11. C - 2.98 g/mL

Density equals mass divided by volume: $44.7 \text{ g} / 15.0 \text{ mL} = 2.98 \text{ g/mL}$.

12. A - 5.96 cm³

$V = m/d = 14.9 / 2.50 = 5.96 \text{ cm}^3$.

13. B - 4.00%

$|24.0 - 25.0| / 25.0 \times 100 = 4.00\%$.

14. D - Accurate on average but not precise

The mean supports accuracy, but the wide spread shows low precision.

15. B - Precise but not accurate

Tight grouping shows precision; consistent displacement from the accepted value shows inaccuracy.

16. A - A systematic offset

A constant zero offset shifts measurements in one direction and is systematic.

17. C - They reveal the spread and support a precision judgment.

Repeated trials show variation and therefore provide evidence about precision.

18. A - It is a defined relationship.

Defined unit relationships are exact rather than measured.

19. D - One additional recorded decimal place

The trailing decimal zero records an additional place of precision; it does not change the magnitude.

20. A - 5.96 cm³

A quantitative result must include its justified value and unit.



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